

Quant II

Lab 3

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2026-02-19

Homeworks

- Homework 1 grades before next recitation

Notes:

- Email to **both** Cyrus & me
- Send **one PDF**, including math, code, and output
 - Use .qmd or .rmd for code+output
 - Math part can be scanned and attached (or from tablet)
LaTeX recommended
- Deadlines are before the lecture

Today's Agenda

- Covariate Adjustment in Experimental Data
- Multiple Regression
- Omitted Variable Bias
- Matching

Covariates for efficiency in randomized experiments

- Lin (2013), fully interacted regression

$$Y_i = \alpha + \rho D_i + (X_i - \bar{X})' \beta_0 + D_i (X_i - \bar{X})' \beta_1 + \epsilon_i$$

- Difference in residualized means: $\hat{\rho} = \bar{\tilde{Y}}_1 - \bar{\tilde{Y}}_0$
- Efficiency gain: $\text{Var}(\tilde{Y}_i(d)) \approx \sigma_{Y(d)}^2 (1 - R_d^2)$
- Intuition: reduce the unexplained variation of the outcome, reduce the standard error on the causal estimate

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- Intuition: reduce the unexplained variation of the outcome, reduce the standard error on the causal estimate
- In small samples instability may arise from issues like covariate imbalance across treatment arms, high leverage, low degree of freedoms
 - Need enough observations **per treatment arm**
 - Prefer a small, pre-specified covariate set strongly predictive of outcome

Implementation with `estimatr::lm_lin`

```
library(fabricatr); library(randomizr)

set.seed(123)

dat <- fabricate(N=1000, x=rnorm(N, mean=2.3),
                x2=rpois(N, lambda = 2),
                x3=runif(N),
                y0=rnorm(N)+x,
                y1=rnorm(N)+x+0.7)

dat$z <- complete_ra(N=nrow(dat))
dat$y <- ifelse(dat$z==1, dat$y1, dat$y0)

# lm_lin
out1 <- estimatr::lm_lin(y~z, covariates=~x, dat)
```


Implementation, alternatives

```
# lm_robust
dat$x_c <- dat$x - mean(dat$x)
out2 <- estimatr::lm_robust(y~z*x_c, dat)

# manually
fit1 <- lm(y ~ x_c, dat[dat$z==1,])
fit0 <- lm(y ~ x_c, dat[dat$z==0,])
rho <- coef(fit1)["(Intercept)"] - coef(fit0)["(Intercept)"]

# compare all
c(coef(out1)["z"], coef(out2)["z"], rho)
```

z	z (Intercept)
0.703241	0.703241

Strong Ignorability Assumptions

Given (Y_i, X_i) , the following allow ρ to be identified:

$$D_i \perp (Y_{1i}, Y_{0i}) \quad \text{and} \quad 0 < \Pr[D_i = 1] < 1 \quad \text{for all } i \in U$$

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or

$$D_i \perp (Y_{1i}, Y_{0i}) \mid X_i \quad \text{and} \quad 0 < \Pr[D_i = 1 \mid X_i = x] < 1 \quad \text{for all } x \in \mathcal{X}.$$

- Informally: “selection on observables”
- Considered strong in most contexts

Identification under CIA

Guarantees that covariate stratum-specific ATEs are identified in the data:

$$\begin{aligned}\tau(x) &= E[Y_{1i} - Y_{0i} | X_i = x] \\ &= E[Y_{1i} | D_i = 1, X_i = x] - E[Y_{0i} | D_i = 0, X_i = x]\end{aligned}$$

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And then τ_{PATE} is identified by averaging over the covariates distribution (cf. Imbens and Wooldridge 2009, p.26-27)

$$\tau_{PATE} = E[\tau(x)] = \int_X \tau(x) dF(x)$$

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Same holds for τ_{PATT} , changing the distribution over which to average:

$$\tau_{PATT} = E[\tau(x) | D_i = 1] = \int_{X|D=1} \tau(x) dF(x | D_i = 1)$$

Multiple regression as an identification strategy

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- Estimates linear approximations of the conditional means of the potential outcomes

Frisch-Waugh-Lovell theorem, a refresher

- Linear model with k covariates. In matrix form: $y = X'\beta + \varepsilon$
- FWL gives a formula for the OLS estimate of the k^{th} coefficient.

$$\hat{\beta}_k = (X'_k M_{[X_{-k}]} X_k)^{-1} X'_k M_{[X_{-k}]} y$$

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- Regress the residuals of y on the residuals for X_k
 - Note: that to get $\hat{\beta}_k$ it is enough to regress the non-residualized y on residualized X_k (why?)
 - but the SE won't be right
- We are often interested in just one regressor (e.g. treatment indicator)

FWL in practice

```
library(tidyverse)
data("mtcars")

# multivariate regression
fit <- lm(mpg ~ cyl + drat + wt, mtcars)

# FWL
resy <- lm(mpg ~ drat + wt, mtcars) %>% residuals()
resx <- lm(cyl ~ drat + wt, mtcars) %>% residuals()
fit2 <- lm(resy ~ resx)

# compare results
out <- c(coefficients(fit)["cyl"], coefficients(fit2)["resx"])
names(out) <- c("Multivariate", "Univariate Residualized")
out
```

Multivariate	Univariate Residualized
-1.509577	-1.509577

Omitted Variable Bias, Intuition

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Decompose both variables into “explained by X” + “residual” pieces:

$$D_i = \widehat{E}[D_i \mid X_i] + \tilde{D}_i, \quad Y_i = \widehat{E}[Y_i \mid X_i] + \tilde{Y}_i.$$

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Because residuals are orthogonal to the regressors, cross-covariances are zero

$$\text{Cov}(Y_i, D_i) = \text{Cov}(\widehat{E}[Y_i | X_i], \widehat{E}[D_i | X_i]) + \text{Cov}(\tilde{Y}_i, \tilde{D}_i)$$

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- FWL/controlling for X removes the first component by construction
- Omitting X leaves it in

Omitted Variable Bias

The coefficient on D_i is $\frac{\text{Cov}(Y_i, D_i)}{\text{Var}(D_i)}$

Omitting X ,

$$\text{Cov}(Y_i, D_i) = \underbrace{\text{Cov}(\rho D_i + X_i' \gamma + v, D_i)}_{\text{Total Variation}}$$

Then,

$$\begin{aligned}\hat{\rho}_{short} &= \rho + \gamma' \cdot \frac{\text{Cov}(X_i, D_i)}{\text{Var}(D_i)} \\ &= \rho + \gamma' \delta\end{aligned}$$

OVB: “Confounder Impact $\times$ Imbalance” (Cinelli & Hazlett 2020)

Sensitivity analysis, i

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- Package website: <https://carloscinelli.com/sensemakr/>

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- Hypothesize different values for them and see if ρ_{true} would cross 0.
- e.g., "If Z is as strong as X , does my effect disappear?"

Sensitivity analysis example, i

Attitudes in Darfur (Hazlett, 2019)

- 2003-2004 government violence against civilians
- Outcome (Y): attitudes toward peace
- Treatment (D): exposure to violence

```
library(sensemakr); library(fixest)
data("darfur")
darfur.model <- feols(
  peacefactor ~ directlyharmed + female +
    age + farmer_dar + herder_dar + pastvoted +
    hhsize_darfur | village,
  data = darfur
)
```


Sensitivity analysis example, ii

```
etable(darfur.model)
```

```

                                darfur.model
Dependent Var.:                peacefactor

directlyharmed    0.0973*** (0.0233)
female            -0.2321*** (0.0237)
age               -0.0021*  (0.0008)
farmer_dar        -0.0404  (0.0292)
herder_dar         0.0143  (0.0321)
pastvoted         -0.0480*  (0.0240)
hhsize_darfur     0.0012  (0.0019)
Fixed-Effects:  -----
village                                Yes
-----
S.E. type                                IID
Observations                                922
R2                                0.32591
Within R2                                0.15423
```

Sensitivity analysis example, iii

```
sensitivity <- sensemakr(darfur.model,  
                        treatment = "directlyharmed",  
                        benchmark_covariates = "female",  
                        kd = 1:4)  
  
# summary(sensitivity)
```

- sensemakr package lets us conduct sensitivity analysis **relative to a covariate of choice**

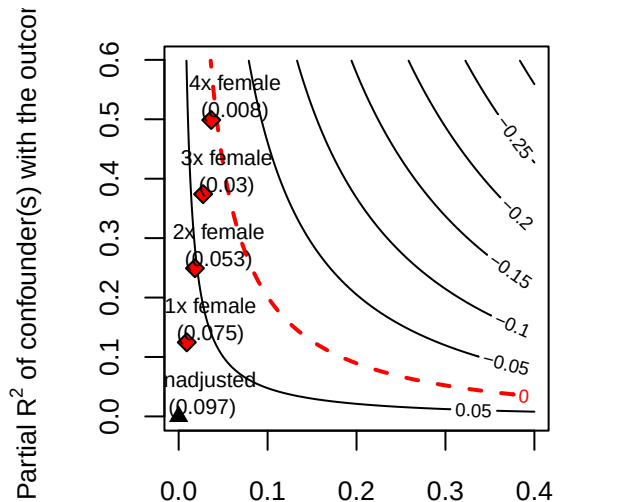
Sensitivity analysis example, iii

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- sensemakr package lets us conduct sensitivity analysis **relative to a covariate of choice**
- kd and ky arguments: a hypothetical confounder **1-4x** as strong as female

Sensitivity analysis example, iv

```
plot(sensitivity)
```



Partial R^2 of confounder(s) with the treatment

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- Not necessarily = the population causal effect.

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When linearity fails

- If we omit interactions, higher order terms
- Specification error

Double machine learning can help (maybe next week?) - but not transparent

Matching

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Then,

- How many units?
- What counts as similar?

Weaker identifying assumptions for ATT

Conditional Mean Independence (CMI)

$$E[Y_{0i}|D_i = 1, X_i] = E[Y_{0i}|D_i = 0, X_i], \quad P(D_i = 1|X_i) < 1$$

- After conditioning on X , treated and control units have the same expected outcome under control.
- Treated units have comparable controls

Exact Matching

Choose matches that have exactly the same value of X_i

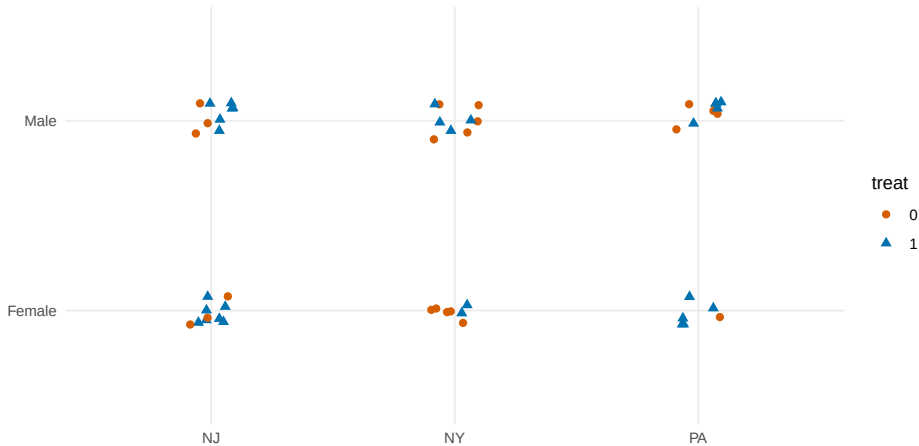
- For each treated unit, find control units with same values of X_i
- Compute the difference in means within these strata and
- For ATT, compute a weighted average using the distribution of X of the treated group

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- For each treated unit, find control units with same values of X_i
- Compute the difference in means within these strata and
- For ATT, compute a weighted average using the distribution of X of the treated group
- Problems
 - Only works with discrete covariates
 - Overlap problems: no control unit available for some levels of X_i

Exact matching, illustration



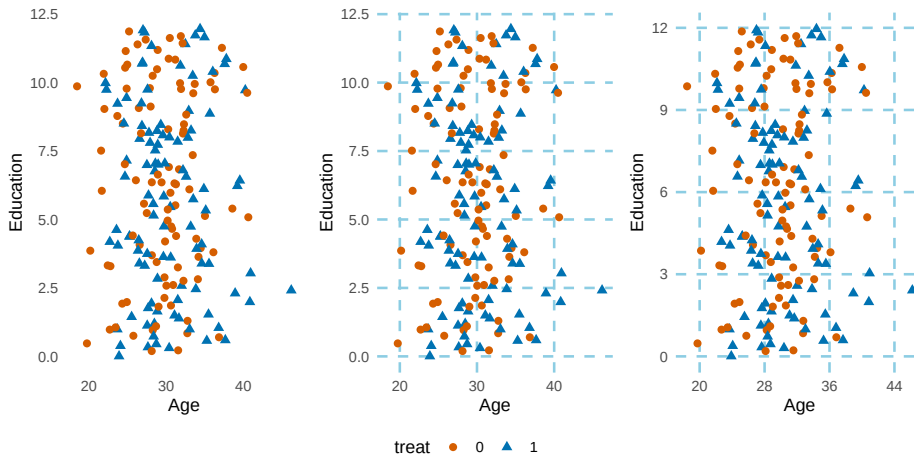
Coarsened Exact Matching

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- Coarsen continuous covariates into discrete bins
- Then exact match on the bin values.
- Problems:
 - Sensitive to bin size
 - Still might be bins with no control units
 - Overlap will determine how narrow the bins can be
- Ideal when you have lots of data and relatively few covariates

Coarsened Exact Matching, illustration



Nearest Neighbor Matching

- Select close units by minimizing some function of the covariates
- Generally create a matched sample of units, and then estimate ATT/ATE
- Researchers can try different matching algorithms until an acceptable level of balance is achieved
 - **Balance Checks**

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 - **Balance Checks**
- Formally for **ATT**:

$$\hat{\tau}_m = \frac{1}{n_1} \sum_{i=1}^n D_i \left(Y_i - \frac{1}{M} \sum_{j \in \mathcal{J}_m(i)} Y_j \right)$$

- $\mathcal{J}(i)$ are the set of M closest control units to i in terms of X_i .
- n_1 number of treated units

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Propensity score:

$$e(X_i) := \Pr[D_i = 1 \mid X_i] = \mathbb{E}[D_i \mid X_i].$$

- Under strong ignorability, $D_i \perp Y_{di} \mid e(X_i)$.
- Matching on $e(X_i)$ is sufficient for identification.

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Then, match using the “nearest neighbors” by this distance metric

NN matching choices

- How many control units per treated?
 - Lower reduces bias
 - More reduces variance

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- How many control units per treated?
 - Lower reduces bias
 - More reduces variance
- With or without replacement: same control matched to multiple treated?
 - **With replacement** minimizes the covariate distance between matched units

NN matching bias correction

- NN matching estimator has generally has asymptotic bias.
- Suppose a treated unit i and matched control j .
- Matching estimate of the treatment effect τ :

$$\begin{aligned}\hat{\tau} &= Y_{i1} - Y_{j0} \\ &= Y_{i1} - Y_{i0} + Y_{i0} - Y_{j0} \\ &= \tau + (Y_{i0} - Y_{j0}).\end{aligned}$$

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Bias corrected

$$\tilde{\tau} = \hat{\tau} - (\hat{Y}_{i0} - \hat{Y}_{j0}) \quad (\text{Abadie and Imbens, 2011}).$$

Evaluating balance

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- QQ plots/KS statistics for comparing the entire distribution of X_i .
- L_1 : multivariate histogram (for CEM)
- Choice of metric can change what matching method works best.

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- Choice of metric can change what matching method works best.

Test for balance

- Do not use simple null hypothesis to test for balance
 - Dropping units can lower power (increase p-values) without a change in balance.
- To test for balance, use an “equivalence test” to see if you could reject some degree of imbalance (Hartman and Hidalgo 2017)

‘MatchIt’ Assessing Balance Page

Matching packages

- MatchIt performs exact matching, genetic, CEM, and NN
 - Different distance metrics allowed
- Other packages: Matching, GenMatch, cem

NN Matching example

Ruggeri, Dorussen, and Gizelis (2016)

```
# Import the data
library(haven)
data <- read_dta("matchingdata.dta")

# Keep non-missing values
data <- na.omit(data)

# Treatment distribution
library(janitor)
tabyl(data$PK0)
```

data\$PK0	n	percent
0	13051	0.94799157
1	716	0.05200843

NN Matching example

```
# Use MatchIt package
library(MatchIt)

# Initial imbalance
nn_raw <- matchit(PKO ~ avgttime + avgadjimr + popgpw2000_40 +
                  avgmnt + borddist + capdist + prec_new,
                  data = data,
                  method = NULL)

knitr::kable(summary(nn_raw)[[3]][, 1:4, drop = FALSE])
```

	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio
distance	0.19	0.04	0.90	5.50
avgttime	317.90	636.58	-1.40	0.35
avgadjimr	140.20	130.98	0.39	0.77
popgpw2000_40	528264.68	184270.83	0.65	2.71
avgmnt	0.19	0.10	0.29	1.97
borddist	141.71	198.89	-0.38	0.97
capdist	579.74	813.60	-0.44	1.56
prec_new	1769.30	1356.78	0.45	2.37

NN Matching example

```
# Nearest-neighbor matching on the propensity score
nn_match <- matchit(PKO ~ avgttime + avgadjimr + popgpw2000_40 +
                    avgmnt + borddist + capdist + prec_new,
                    method = "nearest",
                    data = data)
```

```
nn_match
```

A ``matchit`` object

- method: 1:1 nearest neighbor matching without replacement
- distance: Propensity score
 - estimated with logistic regression
- number of obs.: 13767 (original), 1432 (matched)
- target estimand: ATT
- covariates: avgttime, avgadjimr, popgpw2000_40, avgmnt, borddist, capdist, prec_new

```
# Matched sample
```

```
data_nn <- match.data(nn_match)
```

```
# Matched data
```

```
tabyl(data_nn$PKO)
```

data_nn\$PKO	n	percent
0	716	0.5
1	716	0.5

NN Matching example

```
summary(nn_match)$sum.all[,1:3]
```

	Means Treated	Means Control	Std. Mean Diff.
distance	0.19	0.045	0.90
avgttime	317.90	636.575	-1.40
avgadjimr	140.20	130.981	0.39
popgpw2000_40	528264.68	184270.825	0.65
avgmnt	0.19	0.104	0.29
borddist	141.71	198.893	-0.38
capdist	579.74	813.603	-0.44
prec_new	1769.30	1356.781	0.45

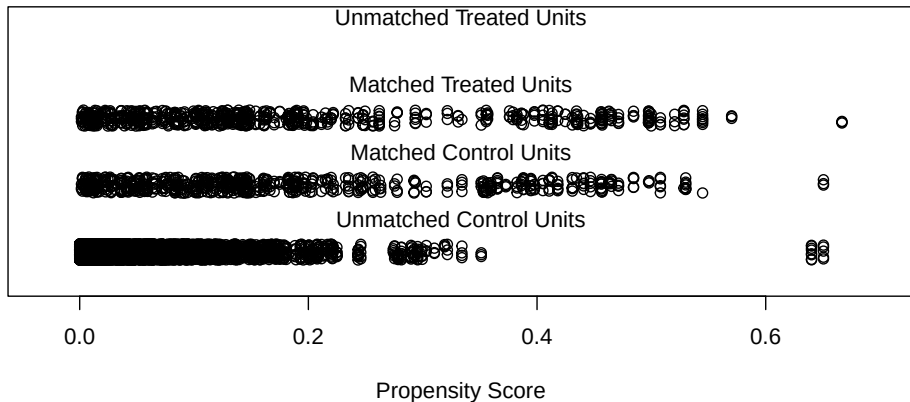
```
summary(nn_match)$sum.matched[,1:3]
```

	Means Treated	Means Control	Std. Mean Diff.
distance	0.19	0.18	0.042
avgttime	317.90	335.22	-0.076
avgadjimr	140.20	145.44	-0.224
popgpw2000_40	528264.68	477351.89	0.096
avgmnt	0.19	0.17	0.060
borddist	141.71	157.15	-0.103
capdist	579.74	504.47	0.140
prec_new	1769.30	1822.87	-0.059

NN Matching example

```
plot(nn_match, type = "jitter", interactive = FALSE)
```

Distribution of Propensity Scores



NN Matching example

```
nn_match_wr <- matchit(PKO ~ avgttime + avgadjimr + popgpw2000_40 +  
  avgmnt + borddist + capdist + prec_new,  
  method = "nearest",  
  replace = TRUE,  
  data = data)  
  
nn_match_wr
```

A `matchit` object

- method: 1:1 nearest neighbor matching with replacement
- distance: Propensity score
 - estimated with logistic regression
- number of obs.: 13767 (original), 931 (matched)
- target estimand: ATT
- covariates: avgttime, avgadjimr, popgpw2000_40, avgmnt, borddist, capdist, prec_

Matched sample

```
data_nn_wr <- match.data(nn_match_wr)
```

Matched data

```
tabyl(data_nn_wr$PKO)
```

data_nn_wr\$PKO	n	percent
0	215	0.23
1	716	0.77

NN Matching example

```
summary(nn_match)$sum.matched[,1:3]
```

	Means Treated	Means Control	Std. Mean Diff.
distance	0.19	0.18	0.042
avgttime	317.90	335.22	-0.076
avgadjimr	140.20	145.44	-0.224
popgpw2000_40	528264.68	477351.89	0.096
avgmnt	0.19	0.17	0.060
borddist	141.71	157.15	-0.103
capdist	579.74	504.47	0.140
prec_new	1769.30	1822.87	-0.059

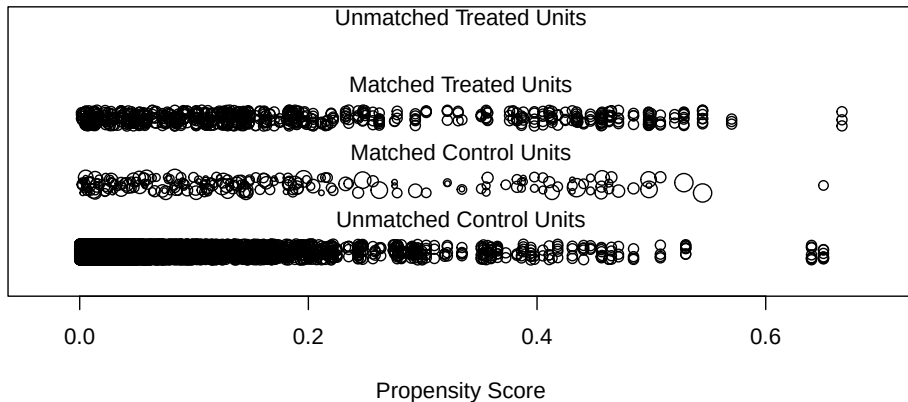
```
summary(nn_match_wr)$sum.matched[,1:3]
```

	Means Treated	Means Control	Std. Mean Diff.
distance	0.19	0.19	0.00078
avgttime	317.90	327.21	-0.04100
avgadjimr	140.20	144.86	-0.19915
popgpw2000_40	528264.68	493444.71	0.06555
avgmnt	0.19	0.20	-0.03328
borddist	141.71	146.11	-0.02943
capdist	579.74	646.31	-0.12409
prec_new	1769.30	1848.36	-0.08704

NN Matching example

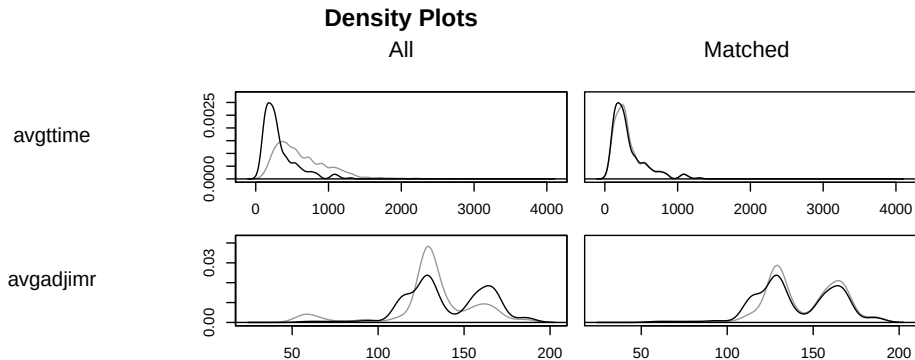
```
plot(nn_match_wr, type = "jitter", interactive = FALSE)
```

Distribution of Propensity Scores



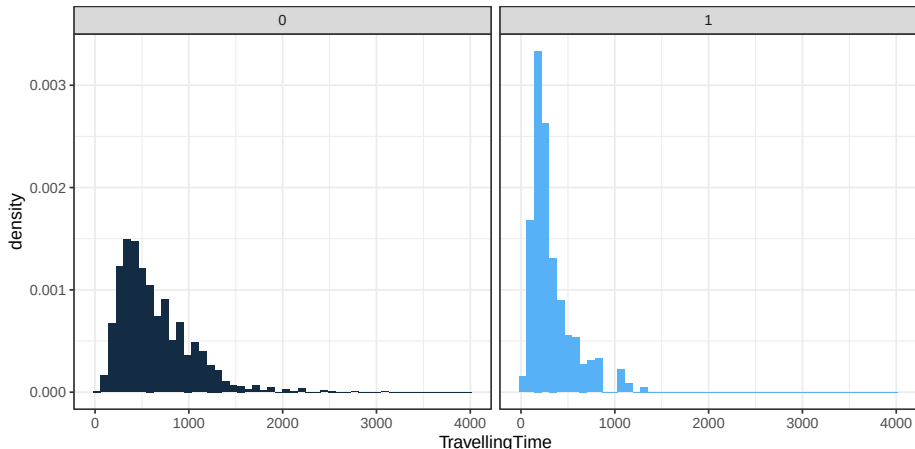
NN Matching example

```
plot(nn_match_wr, type = "density", interactive = FALSE,  
     which.xs = ~avgttime + avgadjimr)
```



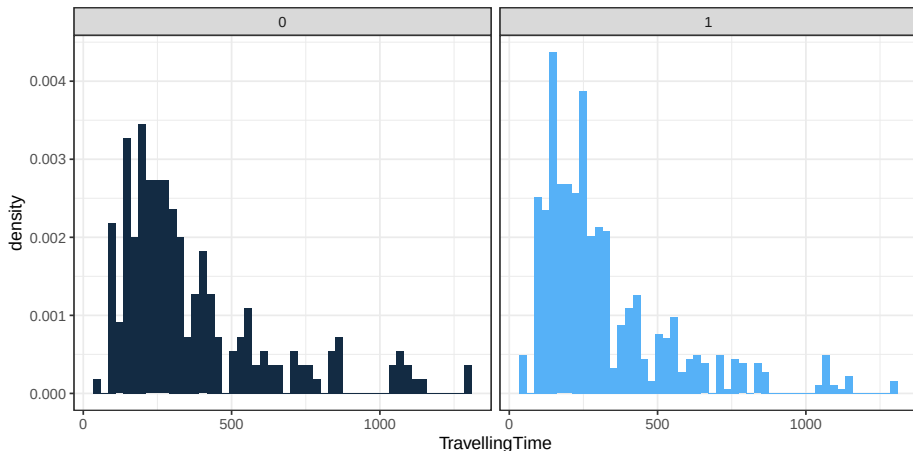
NN Matching example

```
library(ggplot2)
# Raw
ggplot(data, aes(x=avgttime, y = ..density.., fill=PK0)) + geom_histogram(bins=50) +
  facet_wrap(~PK0) + theme_bw() +
  theme(legend.position="none")
```



NN Matching example

```
# matched
ggplot(data_nn_wr, aes(x=avgttime, y = ..density.., fill=PK0)) + geom_histogram(binwidth=50,
  facet_wrap(~PK0)+ theme_bw() +
  theme(legend.position="none")
```



DAGs in R dagitty

```
library(dagitty)

g1 <- dagitty( "dag {
    U -> X -> Y
    X -> D -> Y <- V -> X
}")

children(g1, "X")
```

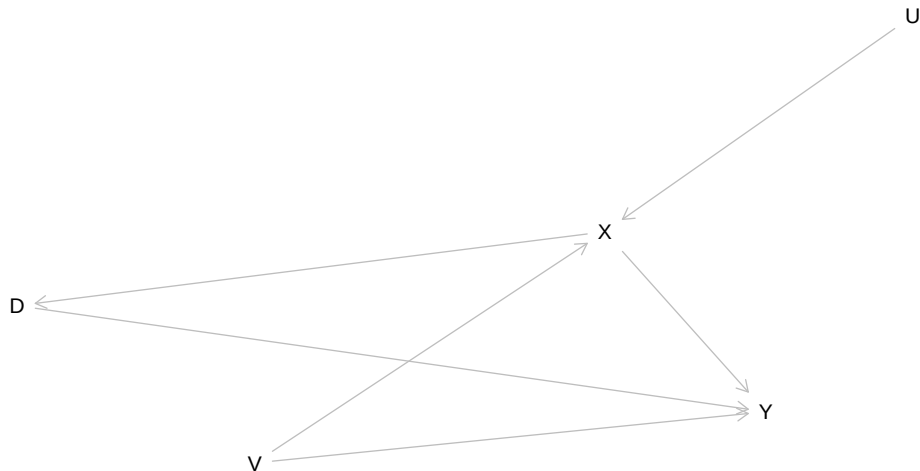
```
[1] "D" "Y"
```

```
parents(g1, "D")
```

```
[1] "X"
```

dagitty

```
plot(graphLayout(g1))
```



```
print( impliedConditionalIndependencies( g1 ) )
```

D _||_ U | X

D _||_ V | X

U _||_ V

U _||_ Y | V, X

```
print( adjustmentSets( g1, "D", "Y", effect="direct" ) )  
  
{ X }
```

dagitty

```
for( n in names(g1) ){  
  for( m in children(g1,n) ){  
    a <- adjustmentSets( g1, n, m, effect="direct" )  
    if( length(a) > 0 ){  
      cat("The coefficient on ",n,"->",m,  
          " is identifiable controlling for:\n",sep="")  
      print( a, prefix=" * " )  
    }  
  }  
}
```

The coefficient on D->Y is identifiable controlling for:
{ X }

The coefficient on U->X is identifiable controlling for:
* {}

The coefficient on V->X is identifiable controlling for:
* {}

The coefficient on V->Y is identifiable controlling for:
{ X }

The coefficient on X->D is identifiable controlling for:
* {}

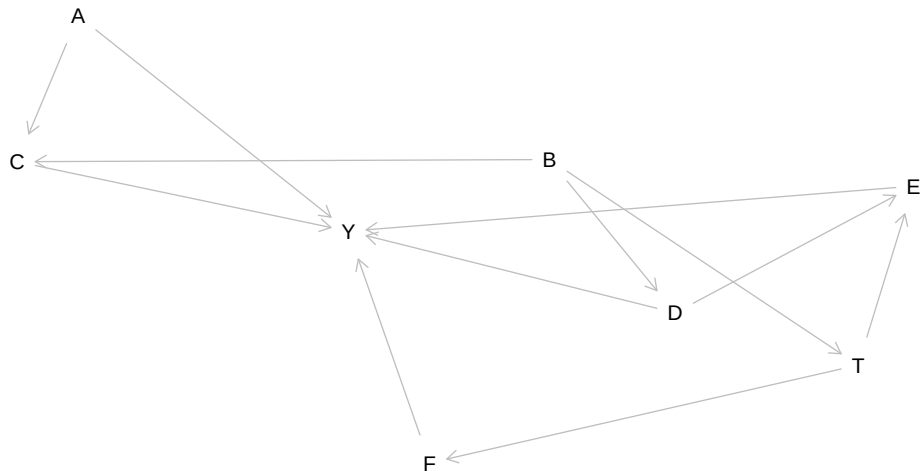
The coefficient on X->Y is identifiable controlling for:
{ D, V }

dagitty

```
g2 <- dagitty( "dag {  
  B -> T -> F -> Y  
  T -> E -> Y  
  B -> D -> E -> Y <- D  
  B -> C -> Y  
  B -> C <- A -> Y  
}")
```



```
plot(graphLayout(g2))
```



```
print(adjustmentSets(g2, "T", "Y", effect="total" ) )
```

```
{ A, C, D }
```

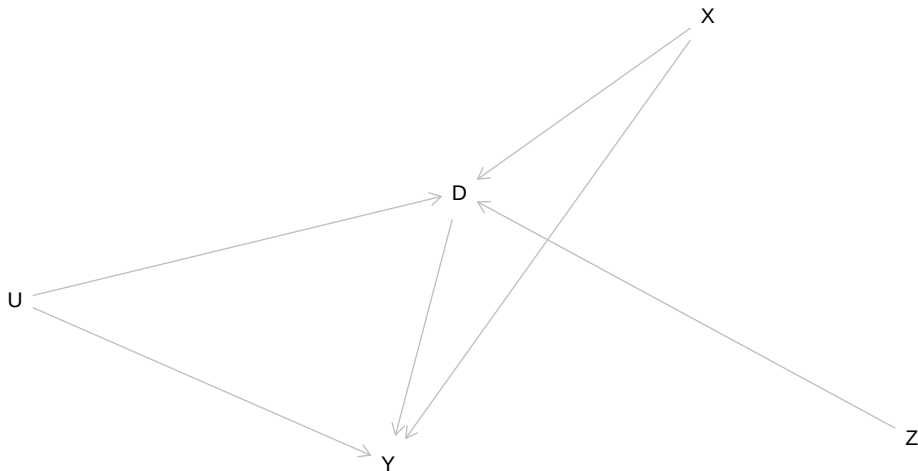
```
{ B }
```

dagitty

```
g3 <- dagitty( "dag {  
    Z -> D -> Y <- X -> D  
    U -> D -> Y <- U  
}")
```

dagitty

```
plot(graphLayout(g3))
```



DAGs in R

```
print(instrumentalVariables(g3, "D", "Y"))
```

Z